

SmartCare v0.1 - Initial Engineering Brief and AI Activity Card

A2 Case Study Stage 1 student resource

SmartCare scenario

SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The client says: 'We need software to help manage patients, practitioners and appointments.' This is not yet a complete specification.

Initial Engineering Brief

1. Problem summary

The clinic wants to switch the manual workflows to digital for ease of use and maintenance. Basically they wanna switch from spreadsheets and paper maintenance to software systems, for ease of creation, updating, deleting, viewing of patient records and appointment records. Also for ease of managing and conflict resolutions of appointments.

2. Initial stakeholders

Stakeholder	What do they need?
Patients	Easy appointment booking, accurate records, privacy, and no clashes.
GP/practitioners	Quick access to patient information and appointment schedules, again no clashes.
Receptionist	Efficient patient registration, searching, displaying, and appointment management (CRUD).
Clinic / IT Devs	Clinic should provide the requirements for what system they need exactly, devs need that PRD to build a correct system.

3. Initial features

Feature	Confirmed or provisional?	Why?
Create a new Appointment Booking.	Confirmed.	Core Feature.
Updating/Deleting Appointment.	Confirmed	Core for conflict resolution.
Viewing/Listing all appointments	Confirmed.	Core for GP to list and see.
Conflict Resolution algorithm.	Provisional	Not yet confirmed, but probably necessary given the nature of system.

4. Questions for the client

1. What sort of data about patients needs to be stored?
2. Who should be allowed to view, edit, delete, create patient records?

3. How are appointments currently booked, resolved in case of conflict, and rescheduled on what basis?
4. Should patients be allowed to view available GP and timeslots?
5. How is the information provided on GP's end? Are they allowed to reschedule appointments too?

5. What we do not yet know

1. We don't know yet about Patient Record management, and GP timeslot management.
2. Don't know yet about how to resolve conflicts if any arises.
3. Don't know if patients are allowed to book online, or only receptionists are able to do so.

AI Activity Card - Ask, Check, Explain

Before AI

What do I think the code does? What problems can I already identify?

I can explain the basic problem statement, and specifications too as stated. I know what the code does and how it does that as explained within the files.

AI request

Act as a tutor. Explain this code and identify potential problems. Do not provide a complete replacement. Ask me questions that help me reason about the solution.

Evaluate

Suggestion	Useful	Unclear	Incorrect	Out of scope
Validate practitioner name and time as well as patient name (for example, reject empty strings).	Yes.			
Treat appointment_time as a real date/time, or at least require a consistent format, so sorting and clash checks are possible.	Yes			
Before append, scan appointments and refuse a booking if that practitioner already has that time.	Yes			

Return the new appointment (or a success message) so the caller knows the booking worked, instead of only mutating a global list.	Yes			
Later (next stage of the course): save the list to a file so bookings survive after the program closes.		Yes, unclear at this stage, probably need some sort of saving mechanism very soon.		

Decide

For each significant suggestion: Accept / Modify / Reject / Keep unverified.

Verify

- Run the code
- Test normal input
- Test unusual input
- Compare with requirements
- Ask tutor/peer
- Check documentation

Explain

Can I explain the final code without reading the AI response? What do I still need to understand?

Yes, I can explain the final code without reading AI response.